Chicago Traffic Crash Risk Scoring

Route Safety Intelligence via CRISP-DM

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Agenda

1. **Business Understanding**
   Problem, stakeholders, goals
2. **Data Understanding**
   Dataset, quality, spatial & temporal patterns
3. **Data Preparation**
   Cleaning, severity weights, feature engineering
4. **Modeling**
   Approach, architecture, validation design
5. **Evaluation**
   Metrics, feature importance, scenario multipliers
6. **Recommendations**
   What to deploy, trade-offs for stakeholders
7. **Conclusion**
   Key findings, limitations, decision takeaway
8. **Future Work**
   Next experiments, data needs, roadmap

The Problem

Core Question

"Given current or selected conditions, which Chicago zones are relatively more dangerous than usual?"

Scale of the problem:

- **USA:** ~6–7M crashes/year; 40,901 fatalities in 2023 (NHTSA)
- **Economic cost:** $340B–$1.4T annually; avg. $50K per crash (FHWA)
- **Illinois 2024:** 303,913 crashes; 1,085 fatal; $15B annual burden
- Chicago alone: 1,045,043 records since 2013 - raw point cloud is unactionable

Stakeholders

- Drivers, cyclists, pedestrians
- City agencies (Vision Zero)
- Logistics & freight operators
- Insurers & navigation platforms

Decision to Support

Rank Chicago grid cells 0–100 under any user-selected scenario so stakeholders can choose safer routes or prioritise infrastructure investment.

Project Goals & Success Criteria

Analytical goals:

- Zone-level risk ranking (relative, not absolute probability)
- Severity-weighted scoring - not just crash frequency
- Condition-switchable layers (rain / dark / wet)
- Temporal generalisation: train 2018–2024, validate 2025

Out of scope:

- Trip-level crash probability
- Causal inference
- Exposure-adjusted safety rates

Three-level Success Criteria

Business: clear zone tiers; switchable layers; interpretable explanations

Analytical: outperform naive hotspot baseline; stable top-K recovery; zero data leakage

MVP: grid CSV ready for map frontend; on-demand scoring API; daily refresh pipeline

Dataset Overview

Three linked CPD tables

Table	Rows	Cols
Crashes	1,045,043	48
Vehicles	2,130,727	71
People	2,293,745	29

Reliable modeling window: 2018–2025 (817,947 rows; pre-2018 coverage insufficient)

Key condition distributions:

- Weather: 78% CLEAR, 8% RAIN, 6% UNKNOWN
- Surface: 73% DRY, 13% WET, 10% UNKNOWN
- Lighting: 64% DAYLIGHT, 22% DARKNESS-LIGHTED

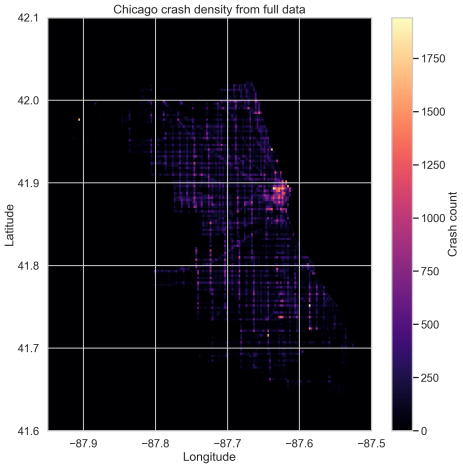
Key quality findings:

- Missing geo-coordinates: only 0.77%
- 4–10% UNKNOWN labels in condition fields - retained as explicit category
- No traffic volume denominator available - results must be relative risk

Main Limitation

Without trip exposure data we cannot compute risk per journey - only risk per zone per period. All scores are comparative, not absolute.

Geographic Crash Distribution

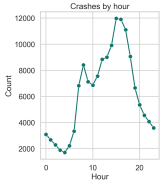


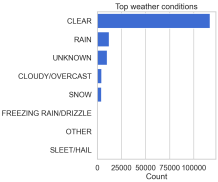
What this shows:

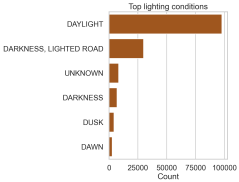
- Clear spatial clustering along arterial corridors
- Dense hotspots on North Side and downtown
- Residential zones have substantially lower raw counts

Business implication:
Raw density predicts future hotspots, but volume alone does not capture *severity* - a low-frequency zone with frequent injuries may deserve higher priority than a high-count minor-collision corridor.

Temporal & Condition Patterns





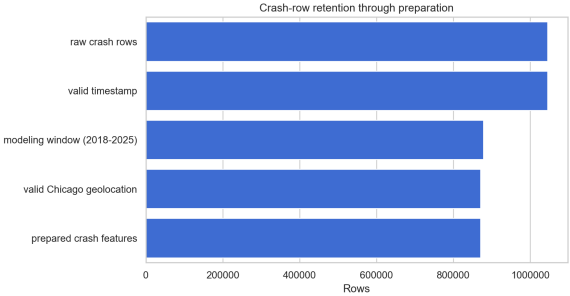


Key patterns:

- Morning and evening rush hours show pronounced peaks
- Rain multiplies crash rates even though clear conditions dominate by volume
- Darkness - even with street lighting - correlates with higher severity outcomes

Business implication:
Condition-aware scoring is essential. A static heatmap misses periods when the same zone becomes far more dangerous.

Cleaning & Filtering



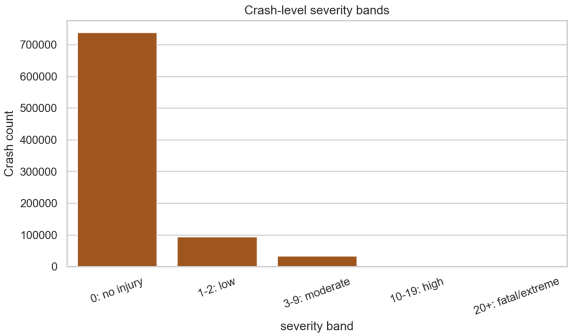
Steps taken:

1. Removed invalid timestamps and out-of-bounds coordinates
2. Normalised categorical fields; encoded UNKNOWN explicitly
3. Excluded post-crash leakage fields from feature set
4. Restricted to reliable window 2018–2025

Why it matters:
Including pre-2018 sparse data or leakage fields would inflate apparent model accuracy while producing unreliable real-world scores.

Final prepared dataset: **870,705 rows**.

Severity Weighting Design

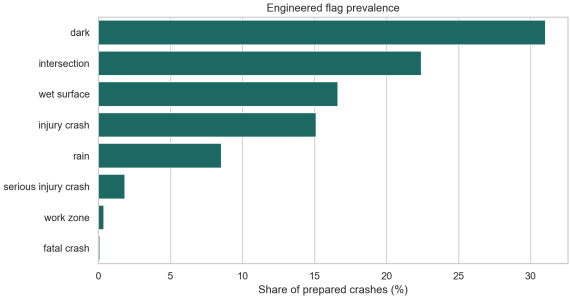


Why weight by severity?
Raw crash counts treat a fender-bender equally with a fatal collision. Severity weights align scores with real-world harm:

Outcome	Weight
Reported, not evident	1
Non-incapacitating	2
Incapacitating	5
Fatal	20

Business implication:
Zones with rare but serious injuries rank higher than zones with many low-severity events - matching stakeholder priorities around injury prevention.

Feature Engineering

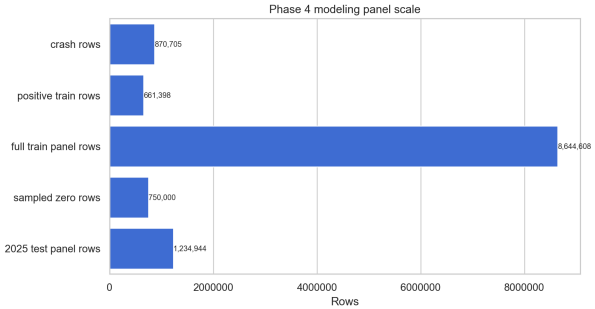


Feature groups built (4 groups):

- **Temporal:** hour, day-of-week, rush hour, night, season, sine/cosine encodings
- **Spatial:** ∼500 m grid cell ID, centroid lat/lon, historical priors (per-grid mean severity, slot mean)
- **Road context:** speed limit, traffic control, trafficway type, intersection flag, work zone
- **Scenario flags:** `is_rain`, `is_dark`, `is_wet_surface`, `is_pedbike`

Total: **48 model features** across 4 groups.

Modeling Approach



Two-layer design:

1. **ML model** - Histogram Gradient Boosting Regressor predicts `log1p(severity_weight_sum)` per grid-time slot
2. **Scenario multipliers** - smoothed condition adjustments applied at query time (no retraining needed)

Why Gradient Boosting?

- Handles mixed numeric/categorical natively
- Robust to sparse grid-time slots
- Fast inference for on-demand scoring
- Interpretable via permutation importance

Model Configuration & Validation Design

Phase 4 model settings

Target	severity_weight_sum on grid_id + crash_year + crash_month + crash_hour + is_weekend
Model	HistGradientBoostingRegressor trained on log1p(severity_weight_sum)
Train/test split	2018-2024 train, 2025 holdout test
Zero-row sampling	full train panel 8,644,608; positives 661,398; sampled zeros 750,000
Weights	positive rows 1.0; sampled zero rows 10.64428
Scenario layer	smoothed grid severity-per-crash multipliers, clipped to stable bounds

Temporal split (anti-leakage):

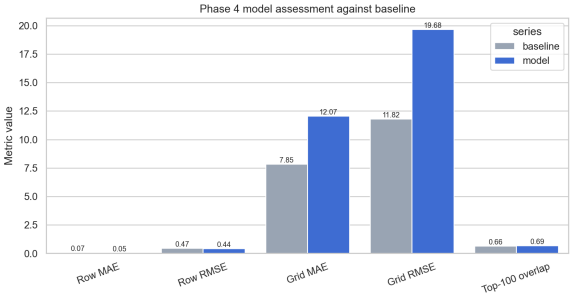
- **Train:** 2018–2024
- **Holdout test:** 2025 (full calendar year)

Grid-time panel construction:

- 8,644,608 total grid-time slots
- 661,398 slots with crashes (8%)
- Zero-slots explicitly included to teach the model what "safe periods" look like
- Zero-slot sampling (750k) with inverse frequency weighting

Key design choice: Zero-filling prevents the model learning only crash-present patterns - critical for reliable low-risk zone identification.

Model Performance vs. Baseline



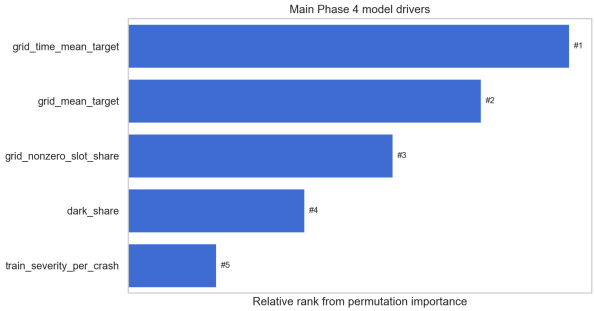
2025 holdout results:

Metric	Baseline	Model
MAE	0.072	**0.054**
RMSE	0.473	**0.444**
R^2	—	**0.35**
Top-100 overlap	0.66	**0.69**

Business interpretation:

- 26% MAE improvement - individual slot predictions meaningfully more accurate
- Top-100 overlap 0.69 means **69 of 100 high-risk zones** correctly predicted
- Zone ranking stable enough for operational use

What Drives Risk - Feature Importance



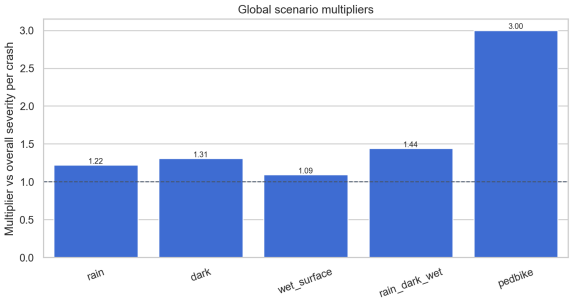
Top drivers:

1. Historical grid-time mean severity
2. Overall grid mean severity
3. Non-zero slot share (crash frequency)
4. Darkness share of the grid
5. Severity-per-crash ratio

Business interpretation:

- History dominates - past dangerous zones remain dangerous; no noise overfitting
- Darkness is a *structural* zone property - zones with poor lighting score higher even in clear conditions
- Severity-per-crash reveals injury-prone zones hidden by low raw crash volume

Condition Scenario Multipliers



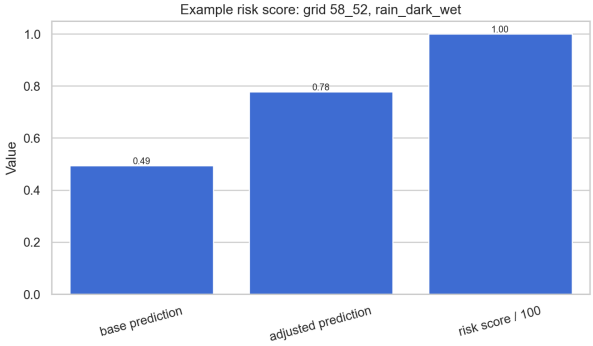
Empirically derived multipliers:

Scenario	Multiplier
Rain only	$\times 1.22$
Darkness only	$\times 1.31$
Wet surface only	$\times 1.09$
Rain + Dark + Wet	$\times 1.44$
Pedestrian / Bike	$\times 3.00$

Business interpretation:

- Combined rain-dark-wet lifts expected severity by **44%**
- Pedestrian/cyclist exposure is **3**$\times$ baseline - critical for active-transport routing
- Scenario switching requires no model retraining - instant response to changing conditions

Example Risk Score Output



Worked example:
Grid 58_52, weekday 18:00, November
Scenario: *rain_dark_wet*

Step	Value
Base ML prediction	0.494
After scenario adj.	0.777
Final 0–100 score	**100**

Business interpretation:

- Score 100 = top-tier risk; avoid or use maximum caution
- Same grid at midday, clear conditions scores substantially lower - conditions matter
- Output is a simple lookup: frontend needs no ML infrastructure

Recommendations for Stakeholders

What to deploy now

- Use precomputed grid scores as backend for interactive Chicago safety map
- Enable six scenario layers: overall, rain, dark, wet surface, combined, ped/bike
- Surface top-5 risk drivers per zone so users understand *why* a zone scores high
- Prioritise ped/bike mode for cycling-route and pedestrian-safety applications

Trade-offs to communicate

- Scores are **relative**, not probabilities - score of 80 $\neq$ 80% crash chance
- Historical priors dominate - new roads or infrastructure changes take time to be reflected
- $\sim$500 m resolution may conflate safe and dangerous micro-locations

Priority action

Connect a live weather API to auto-select scenarios so the map reflects current conditions without manual user input.

Conclusion

All success criteria met:

- ✓ 2,144 grid cells ranked 0–100
- ✓ Six deployable scenario layers
- ✓ 26% MAE improvement over naive baseline
- ✓ 0.69 top-100 hotspot overlap on unseen 2025 data
- ✓ Strict temporal split - zero leakage
- ✓ Flat-file outputs ready for frontend
- ✓ Interpretable feature drivers

Known limitations:

- No traffic-volume denominator - relative risk only
- 4–10% UNKNOWN condition labels dilute scenario signal
- Fixed grid resolution may hide within-cell variation
- No confidence intervals on scores

Decision takeaway

Model is production-ready as a relative risk-ranking backend.

Deploy for: route safety guidance, infrastructure prioritisation (Vision Zero), condition-aware alerts.

Treat scores as **comparative tiers**, not absolute probabilities.

Future Work

Near-term improvements:

- Incorporate vehicle and people context features (prepared but not yet in model)
- Rolling cross-validation within training window
- Systematic hyperparameter search
- Quantile regression for prediction intervals

Data enhancements:

- Live weather API for auto scenario selection
- Traffic exposure proxies (AADT, transit ridership)
- Road-segment-level granularity
- Real-time traffic feeds

Operational roadmap (Phase 6):

1. Build interactive map frontend
2. Automate data refresh and monthly retraining
3. Expose lightweight scoring API
4. Add crash-trend alerting for zones showing increasing severity
5. Causal policy evaluation - measure impact of infrastructure changes on scores

Confidence-building priority

Adding prediction intervals and exposure denominators would allow stakeholders to act on scores with higher statistical confidence and support formal policy decisions.

Thank You

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All six CRISP-DM phases complete. Project ready for deployment.

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